

Advanced Data Analysis

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Chapter 1

Prerequisites

```
install.packages("bookdown")  
# or the development version  
# devtools::install_github("rstudio/bookdown")
```


Chapter 2

Introduction

3 types of data

- Structured Data (tables form)
- Semi-structured Data (e.g., XML, JSON, HTML)
- Unstructured Data
 - Texts
 - Images: Videos, photos,
 - Networks: Social Networks

Chapter 3

Machine Learning

This section based on notes in “Analyzing Unstructured Data” course taught by professor Kunpeng Zhang.

- ML is concerned with the the development, the analysis, and the application of algorithm that allow computers to learn
- Learning:
 - A computer learns if it improves its performance at some tasks with experience (i.e., data)
 - Extracting a model of system from the sole observation (or the simulation)
 - A model = some relationships between the variables used to describe the system.
- Goals:
 - Make prediction
 - Understand the underlying mechanisms.
- Learning if useful when:
 - Human expertise does not exist (navigating on Mars)
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing on a computer network)
Solution needs to be adapted to particular cases (user biometrics)
- Applications in different domains:

- Retail: Market basket analysis, customer relationship management (CRM)
- Finance: Credit scoring, fraud detection.
- Manufacturing: Optimization, troubleshooting
- Medicine: medical diagnosis
- Telecommunications: Quality of service optimization, routing
- Bioinformatics: Motifs, alignment WEb mining, search engines.
- Steps
 - Data generation: data types, sample size
 - Preprocessing: normalization, missing values, feature selection/extraction
 - Machine Learning: Hypothesis, choice of learning paradigm/algorithm.
 - Hypothesis validation: cross-validation, model deployment.
- Types of machine learning:
 - Supervised: manually label variables
 - Unsupervised:
- Training and testing
 - Training is the process of making the machine learn
 - No free lunch:
 - * Training set and testing set come from the same distribution
 - * Need to make some assumptions.
- Factors affecting the performance:
 - Types of training provided
 - The form and extend of prior knowledge
 - Type of feedback provided
 - The learning algorithms used
- Two factors:

- Modeling
- optimization
- Algorithms:
 - ML depends on algorithms.
 - The algorithms control the search to find and build the knowledge structures
 - The learning algorithms should extract useful information from training examples.
- Types of learning (algorithms):
 - Supervised learning ($\{X_n \in R^d, y_n \in R\}_{n=1}^N$)
 - * Prediction
 - * Classification (discrete labels), regression (real values)
 - Unsupervised learning ($\{X_n \in R^d\}_{n=1}^N$)
 - * Clustering
 - * Probability distribution estimation
 - * Finding association (in features)
 - * Dimension reduction
 - Semi-supervised learning
 - Reinforcement learning
 - * Decision making (robot, chess machine)

3.1 Supervised Learning

From the data, find a function f of the inputs that best approximates the outputs.

$$\hat{Y} = f(X_1, \dots, X_n)$$

- Predictive: Make predictions for a new object described by its attributes
- Informative: help understand the relationship between inputs and outputs

Divide data into

- Training set: train model
- validation set: tune hyper parameters
- Test set: evaluation of the model for generalization error.

Learning algorithm

- defined by
 - a family of candidate models (= **hypothesis space H**)
 - a quality measure for a model
 - an optimization strategy
- takes as input a learning sample and outputs a function h in H of maximum quality

Model selection

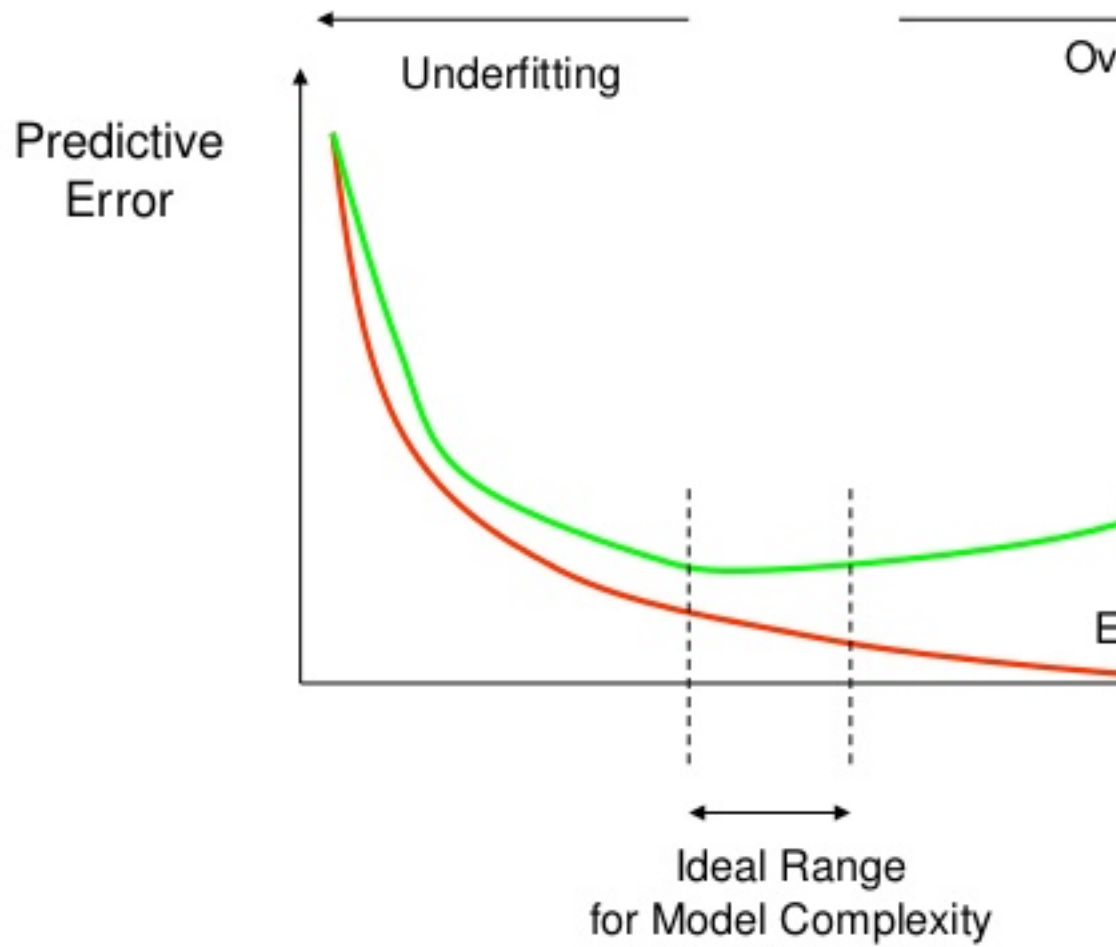
- Why not choose the model that minimizes the error rate on the learning sample? (**re-substitution error**)
- How well are you going to predict future data drawn from the same distribution? (**generalization error**)
- Evaluation on test set
 1. Randomly choose a percentage (e.g., 30%) of the data to be in a test set
 2. The remainder is training set
 3. Learn the model from the training set
 4. Estimate its future performance on the test set.
- Problems:
 - overfits: occurs when the learning algorithm starts fitting noise.
 - unfits
- Evaluation on test set:
 - Good:
 - * Very simple
 - * Computationally efficient
 - Bad:

- * Waste data
- * very unstable when database is small
- Another evaluation: **Leave-one-out cross validation**
 - For $k = 1$ to N
 - * remove the k -th object from the learning sample
 - * learn the model on the remaining objects
 - * apply the model to get a prediction for the k -th object
 - report the proportion of mis-classified objects
- Good:
 - Do not waste the data (you get an estimate of the best method to apply to $N-1$ data)
- Bad:
 - Computationally intensive (train N models)
 - high variance.
- Another evaluation: **K-fold cross validation**
 - randomly partition the dataset into k subsets (e.g., 10)
 - For each subset:
 - * learn the model on the objects that are not in the subset
 - * compute the error rate on the points in the subset. Report the mean error rate over the k subsets.
 - When $k =$ the number of objects $\rightarrow$ leave-one-out-cross validation.

Rule-of-thumb for evaluation

- Lots of data (>1000) test set validation
- small data (100-1000): 10-fold CV
- very small data (<100) leave one out CV.

How Overfitting affects Prediction



Use the testing set to estimate the performance of the final model

Performance measures

- The error rate is no the only way to assess a predictive model.
- In binary classification, results can be summarized in a contingency table (i.e., confusion matrix)

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Various criteria

- Error

ROC and precision/recall curves

- Each point corresponds to a particular choice of the decision threshold.

Also known as robustness check

You want large area under the curve

Comparison of learning models

- Three main criteria:
 - Accuracy: Measures by the generalization error (estimated by CV)
 - Efficiency: Computing times and scalability for learning and testing
 - Interpretability: Comprehension brought by the model about the input-output relationship.
- Depends on the application, you have to make trade-off.

Learning Techniques

- Supervised learning categories and techniques
 - Linear classifier (numerical functions):
 - * Perceptron
 - * Logistic regression
 - * Support vector machine (SVM): linear to nonlinear: feature transform and kernel function
 - * Ada-line
 - * Multi-layer perceptron (MLP)
 - Parametric (Probabilistic functions)
 - * Naive Bayes, Gaussian discriminant analysis (GDA), Hidden Markov models (HMM), probabilistic graphical models
 - Non-parametric (Instance-based functions)
 - * K-nearest neighbors, Kernel regression, Kernel density estimation, local regression.
 - Non-metric (symbolic functions)
 - * Classification and regression tree (CART), decision tree
 - Aggregation
 - * Bagging (bootstrap + aggregation), Adaboost, Random forest.
- Unsupervised learning categories and techniques:
 - Clustering
 - * K-means clustering
 - * Spectral clustering
 - Density Estimation
 - * Gaussian mixture model (GMM)

- * Graphical models
- Dimensionality reduction
 - * Principal component analysis (PCA)
 - * Factor analysis
- Semi-supervised learning
 - goal: exploit both labeled and unlabeled data to build better models than using each one alone
 - Self-training:
 - * Iteratively label some unlabeled examples with a model learned from the previously labeled examples
 - * Active learning.
- Reinforcement learning
 - Learning from interactions
 - Goal: learn to choose sequence of actions (=policy) that maximizes $r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$ where $0 \leq \gamma < 1$

3.2 Unsupervised Learning

3.2.1 Clustering

Clustering is used to find similar groups in data (i.e., clusters)

No class value denoting an a priori grouping (hence, Unsupervised Learning)

Clustering algorithm

- Partitional clustering
- Hierarchical clustering

A distance (similarity, or dissimilarity) function

Clustering quality

- Inter-clusters distance -> maximized
- Intra-clusters distance -> minimized

The quality of a clustering result depends on

- The algorithm

- the distance function
- the application

3.2.1.1 Similarity Measures

It's your choice of how to define similarity

(Dis)similarity measure

- Equivalently, we can use dissimilarity measures
- Jagota defines a dissimilarity measure as a function $f(x,y)$ such that $f(x,y) > f(w,z)$ if and only if x is less similar to y than w is to z .
Also known as **pair-wise** measure

Distance functions

Different distance functions for

- Different types of data
 - Numeric data
 - Nominal data
- Different applications

Numeric data

- $dist(x_i, x_j)$, where
 - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + \dots + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
 - x_i and x_j are data points (vectors)
- Euclidean distance
 - If $h = 2$, it is the Euclidean distance: $dist(x_j, x_j) = ((x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
 - Weighted Euclidean distance: $dist(x_j, x_j) = (w_1(x_{i1} - x_{j1})^2 + w_2(x_{i2} - x_{j2})^2 + \dots + w_r(x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
 - Squared Euclidean distance: to place progressively greater weight on data points that are further apart: $dist(x_j, x_j) = (x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ir} - x_{jr})^2$
- Manhattan distance
 - If $h = 1$, it is the Manhattan distance: $dist(x_j, x_j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ir} - x_{jr}|$
- Minkowski distance (h is positive integer)
 - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + \dots + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
- Chebychev distance: define 2 data points as “different” if they are different on any one of the attributes

$$- \text{dist}(x_j, x_j) = \max(|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ir} - x_{jr}|)$$

Binary and nominal Attributes

- Binary attribute: has 2 values or states but no ordering relationship (e.g., gender)
- We use a confusion matrix to introduce the distance function/measures
- Let the i-th and j-th data points be X_i and x_j (vectors)

3.2.1.2 K-mean Algorithm

Chapter 4

Methods

We describe our methods in this chapter.

Chapter 5

Applications

Some *significant* applications are demonstrated in this chapter.

5.1 Example one

5.2 Example two

Chapter 6

Final Words

We have finished a nice book.

Part I

ADVANCED

Chapter 7

Deep Learning

7.1 Overview

- What is deep learning? Input to output via layers of representation
- What are layers? A layer is a geometric transformation function on the data that goes through it Weights determine the data transformation behavior of a layer

Learning Representation

- Transforming input data into useful representation

The “deep” in deep learning

- multiple layers
- Other possibly more appropriate names for the field:
 - Layered representations learning
 - Hierarchical representations learning
 - Chained geometric transformation learning

New problem domains for R:

- Computer vision
- Computer speech recognition
- Reinforcement learning applications

How do we train deep learning models?

- Basic of machine learning algorithms

- Machine learning vs. statistical modeling
- MNIST example
 - Model definition in R
 - Layers of representation
- The training loop

Machine learning algorithms

Learning model parameters via exposure to many example data points

Statistics: Often focused on inferring the process by which data is generated

Machine Learning: Principally focused on predicting future data

Deep learning frontiers

- Computer vision
- Natural language processing
- Time series
- Biomedical

7.2 Tensor Flow and R

TensorFlow APIs

- Keras API
- Estimator API
- Core API

7.2.1 R packages

TensorFlow APIs * `keras`: Interface for neural networks, with a focus on enabling fast experimentation. * `tfe`: Implementations of common model types such as regressors and classifiers.

* `tensorflow`: Low *level interface to the TensorFlow computational graph*. `tf-datasets`: Scalable input pipelines for TensorFlow models.

Supporting Tools * `tfruns`: Track, visualize, and manage TensorFlow training runs and experiments * `tfdeploy`: Tools designed to make exporting and serving TensorFlow models straightforward. * `cloudml`: R interface to Google Cloud Machine Learning Engine.

R interface to Keras

- Step by step example
- Keras layers
- Compiling models
- Losses, Optimizers, and Metrics
- More examples

7.2.2 Keras layers

65 layers available

- Dense layers: classic “fully connected” neural network layers
- Convolutional layers: “Filters for learning local patterns in data
- Recurrent layers: Layers that maintain state based on on previously seen data
- Embedding layers: Vectorization of text that reflects semantic relationships between words

7.3 Compiling models

Model compilation prepares the model for training by:

1. Converting the layers into a TensorFlow graph
2. Applying the specified loss function and optimizer
3. Arranging for the collection of metrics during training

7.4 Losses, Optimizers, and Metrics

All available at Keras for R cheatsheet

Example of Image classification

7.5 NYU

Materials in this chapter is based on NYU Deep Learning

Deep in Deep Learning means multi-layers. “A deep network has several layers and uses them to build a hierarchy of features of increasing complexity”

Supervised Learning (= Function Optimization): With inputs and outputs, you train the machine to tweak its parameter to have the correct outputs from inputs. In “traditional machine learning”, you have to engineer **feature extractor**, but in deep learning you can multi-layers feature extractor can be trained. And all layers are non-linear, because if they are linear, then all layers would be collapse into one layer.

Natural Data is compositional. Hence, it is efficiently representable hierarchically.

Revolutions in Deep Learning:

- Speech Recognition: 2010
- Image Recognition: 2013
- Natural Language Processing (NLP): 2015

Deep Learning = Learning Representations/ Features Representation means you turn raw data into something useful

Image Recognition:

Pixel, edge, texture, motif, part, object

Text

Character, group, word group, clause, sentence, story

Speech

Sample, spectral band, sound, phone, phoneme, word.

Difference between Deep Learning Support-Vector Machines (SVM)

SVM is a 2-layer neural net, where

- layer 1 = templates
- layer 2 = linear classifier.

where Deep Learning is “A deep network has several layers and uses them to build a hierarchy of features of increasing complexity”

Deep Machines are more efficient than SVM in representing certain classes of functions.

The Manifold Hypothesis “states that real-world high-dimensional data lie on low-dimensional manifolds embedded within the high-dimensional space”

Difference between Deep Learning and PCA (Principal Components Analysis) is that Deep learning uses non-linear function, while PCA uses linear function.

Chapter 8

Interpretable Machine Learning

<https://christophm.github.io/interpretable-ml-book/> https://en.wikipedia.org/wiki/Explainable_artificial_intelligence <https://arxiv.org/pdf/1905.08883.pdf>
<https://arxiv.org/abs/2004.14545> <https://www.kaggle.com/learn/machine-learning-explainability> <https://towardsdatascience.com/interperable-vs-explainable-machine-learning-1fa525e12f48>

8.1 SHAP

<https://github.com/slundberg/shap> <https://meichenlu.com/2018-11-10-SHAP-explainable-machine-learning/> <https://bjlkeng.github.io/posts/model-explanability-with-shapley-additive-explanations-shap/>

8.2 Video Classification

<https://hackernoon.com/continuous-video-classification-with-tensorflow-inception-and-recurrent-nets-250ba9ff6b85> <https://cloud.google.com/video-intelligence> <https://arxiv.org/pdf/1609.06782.pdf> <https://www.tubefilter.com/2019/02/12/taxonomy-youtube-videos-develop-original-content-that-works/>
<https://medium.com/swlh/converting-news-into-video-stories-5d8d4fc5a32b>
<https://towardsdatascience.com/introduction-to-video-classification-6c6acbc57356>
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0228579>

8.3 Image Memorability

(AMNET)

https://www.researchgate.net/publication/335603628_Image_Memorability_Using_Diverse_Visual_Features_and_Soft_Attention https://www.researchgate.net/publication/308812354_Deep_Learning_for_Image_Memorability_Prediction_the_Emotional_Bias <https://arxiv.org/pdf/2005.02969.pdf> <https://github.com/dazcona/memorability> <https://arxiv.org/abs/1804.03115> https://www.researchgate.net/publication/324387176_AMNet_Memorability_Estimation_with_Attention http://ceur-ws.org/Vol-2283/MediaEval_18_paper_24.pdf <https://jov.arvojournals.org/article.aspx?articleid=2771173> https://openaccess.thecvf.com/content_cvpr_2018/papers/Fajtl_AMNet_Memorability_Estimation_CVPR_2018_paper.pdf <https://ieeexplore.ieee.org/document/8578764>
<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7382237&tag=1> <https://people.umass.edu/~yangzhou/>
<http://people.csail.mit.edu/khosla/projects/regionmem/> <https://github.com/adikhosla/feature-extraction> <https://expertprogrammanagement.com/2019/12/elaboration-likelihood-model/>

Chapter 9

Research Methods

9.1 But-for World

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Mahajan, V., Sharma, S., and Buzzell, R. D. (1993). Assessing the impact of competitive entry on market expansion and incumbent sales. *Journal of Marketing*, 57(3):39.